

Research paper

Objectively measured differences in physical activity in five types of schoolyard area



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HIGHLIGHTS

- We investigate how schoolchildren use different schoolyard areas during recess.
- We examine schoolyard behavior using GPS, accelerometer and GIS.
- Grass and playground areas had the highest proportion of moderate to vigorous activity.
- Solid surface areas had the highest proportion of time spent sedentary.
- Girls accumulated more sedentary time in all area types compared to boys.

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ABSTRACT

Physical activity (PA) in childhood is related to a multitude of short- and long-term health consequences. School recess can contribute with up to 40% of the recommended 60 min of daily moderate-to-vigorous physical activity (MVPA). This paper aims to investigate how schoolchildren use different schoolyard areas during recess and whether these areas are associated with different levels of PA. Time spent by 316 students (grade 5–8) in five types of schoolyard area was measured during at least two days and four separate recess period per person (in total 1784 recess periods), using global positioning system (GPS) and the level of activity was measured using accelerometers. Total time spent and proportions of time spent sedentary and in MVPA were calculated per area type. Significant differences in PA levels were found. Grass and playground areas had the highest proportion of time in MVPA and solid surface areas had the highest proportion of time spent sedentary. Boys and children spent a higher proportion of time in MVPA. Girls accumulated more sedentary time in all area types compared to boys. This finding emphasizes the importance of investigating various settings and features in schoolyards in promoting PA. Grass and playground areas may play an important role in promoting PA in schoolyards, while a high proportion of time in solid surface areas is spent sedentary. In future, more detailed studies of the exact schoolyard setting using a combination of GPS, accelerometer and direct observation would be beneficial.

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1. Introduction

Physical activity (PA) in childhood is related to a multitude of short- and long-term health consequences by its preventive effect on physical conditions, such as obesity (Harrison & Jones, 2012; Mark & Janssen, 2009), markers of cardio metabolic risk factors (Holman, Carson, & Janssen, 2011), low bone mass and increased blood pressure and cholesterol (Janssen & Leblanc, 2010). PA has also proved to be crucial to the cognitive performance (Sibley & Etnier, 2003) and mental well-being of the child, by preventing the development of depression (Janssen & Leblanc, 2010) and

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strengthen the self-esteem (Ekelund et al., 2004). A systematic review indicates that the more time children and youth engage in PA of at least moderate intensity, the greater the health benefits they achieve (Janssen & Leblanc, 2010).

The importance of promoting PA in childhood is confirmed by the fact that the PA level early in life seems to follow into adulthood (Pate, Baranowski, Dowda, & Trost, 1996; Telama et al., 2005).

Despite the current public health recommendation to participate in 60 min or more of moderate-to-vigorous physical activity (MVPA) each day, a study from the United States shows that 42% of the children aged 6–11 years old, and only 8% of the adolescent aged 12–19 years old, meet the recommendations (Troiano et al., 2008). A report from the Danish national board of health indicates that 58% of the boys and 55% of the girls aged 11–15 years old are physically active at least 60 min a day (Sundhedsstyrelsen, 2006). Additionally, the pattern of children's sedentary behavior is a concern. Several studies indicate that children and adolescents spent a considerable proportion of their day sedentary, with sustained sequences of 20–30 min or more (Abbott, Straker, & Erik Mathiassen, 2013; Harrington, Dowd, Bourke, & Donnelly, 2011). School time in particular has shown to be associated with more uninterrupted sedentary periods than non-school time (Abbott et al., 2013). A study by Biddle, Pearson, Ross, and Braithwaite (2010) found that sedentary behavior patterns established during childhood or adolescence often are followed into adulthood, and that this so-called tracking seems to be stronger for sedentary behavior than for PA. In adults, sedentary time has shown to be an independent risk factor for obesity and diabetes (Katzmarzyk, 2010). Sedentary time may as well have detrimental effect on children's health if the findings observed in adults are true for children (Healy et al., 2008).

The school is recognized as an important setting for promoting PA and reducing sedentary time due the major role it plays in children's and adolescent's life (Mota et al., 2008; Naylor & McKay, 2009). Within the school, two main contexts exist in which children have the opportunity to be physically active; the physical education (PE) lessons and recess. A review by Biddle and Asare (2011) found that recess PA improves cognitive performance; academic achievement; classroom behavior, attention and concentration. Several other studies have also demonstrated the importance of recess in the promotion of PA (McKenzie, Crespo, Baquero, & Elder, 2010; Ridgers, Fairclough, & Stratton, 2010; Ridgers, Saint-Maurice, Welk, Shiahpush, & Huberty, 2011). A study by Ridgers, Stratton, and Fairclough (2006) found that activity during recess can contribute with up to 40% of the daily recommendation of 60 min of MVPA. Gender differences in recess PA are well documented, with boys typically engaging in more PA and girls spending more time sedentary (Bailey et al., 2012; Hilland, Ridgers, Stratton, & Fairclough, 2011; Ridgers, Salmon, Parrish, Stanley, & Okely, 2012; Ridgers, Stratton, & Fairclough, 2005). In addition, PA during recess was found to decrease with age (Fjortoft, Kristoffersen, & Sageie, 2009; Fjortoft, Lofman, & Halvorsen Thoren, 2010). Apart from these demographic factors, various other determinants have shown to influence PA during recess including biological (Ekelund et al., 2004; Hilland et al., 2011) and physical environmental factors (Nielsen, Bugge, Hermansen, Svensson, & Andersen, 2012; Taylor et al., 2011). Time spent outdoors also has a potential influence on children's PA. Previous studies have found that objectively measured PA was 2–3 fold higher outdoors than indoors (Cooper et al., 2010; Ferreira et al., 2007). From a health promotion perspective, it is interesting particularly to look at the physical environment factors, as many of these can be changed relatively easy by changing the design and contents of a schoolyard.

Several studies have described the impact of schoolyard characteristics on PA and found various factors to be conducive to PA such as playground area per student (Fairclough, Beighle, Erwin, &

Ridgers, 2012), playgrounds colors (Ridgers, Stratton, Fairclough, & Twisk, 2007; Stratton & Mullan, 2005) and the amount of removable and permanent equipment available (Haug, Torsheim, Sallis, & Samdal, 2010; Nielsen, Taylor, Williams, & Mann, 2010; Zask, van Beurden, Barnett, Brooks, & Dietrich, 2001).

The majority of these studies have used observations or self-reported measurements when studying the association between schoolyard characteristics and children's movement patterns (Hilland et al., 2011; Ridgers et al., 2010; Taylor et al., 2011). Only few studies used objective measures (Global positioning systems (GPS) in combination with heart rate monitor), in assessing the movement patterns and PA levels of children aged 6 and 14 years old in one recess (Fjortoft et al., 2009, 2010). More research is needed to identify environmental indicators for PA in children at different gender and age groups according to one of these studies (Fjortoft et al., 2009).

The aim of this study was to investigate the use of different schoolyard areas and determine if these settings were related to differences in PA and sedentary behavior in schoolchildren (students) attending grade 5–8 using GPS and accelerometer. The objective will be addressed through the following specific research questions:

- How much time do students spend in the different schoolyard areas?
- What is the proportion of time spent in MVPA and sedentary out of the time spent in each schoolyard area?
- Are there gender or age differences in use and activity level in the different schoolyard areas?

2. Material and methods

2.1. Population

This study used baseline data from the When Cities Move Children (WCMC) study. WCMC is a longitudinal natural experiment conducted in and around the Haraldsgade district in Copenhagen, the capital of Denmark. The aim of the WCMC study is to document and assess the impact of urban renewal in the Haraldsgade district on children's PA and urban movement. Four schools in the district were invited to participate in the WCMC study (Klinker, Schipperijn, Kerr, Ersboll, & Troelsen, 2014b).

The district in which the four schools are located is characterized by being very multi-ethnic with 30 different nationalities represented. Children comprise 20% of the roughly 9300 people living in the district, and about 70% of these children are categorized by Statistics Denmark as immigrants or descendants of immigrants (Klinker et al., 2014a).

On behalf of the parents, the school boards of the four schools granted permission to include all their students in grade 5–8 (aged 10 to 15 years old) in the study. The parents received a passive informed consent form which the students could return if they did not want to participate, and they could withdraw during any stage of the study. The data were collected between April 2010 and September 2011. At the time of the study, 623 students attended the four schools in grade 5–8, from which 523 participated in the study (an 84% response rate).

2.2. Instruments

PA was recorded as an activity-count every 2 s using the ActiGraph accelerometer model GT3X. The ActiGraph accelerometer is a small lightweight activity monitor designed to detect movements in vertical, horizontal and transverse axes. But since the validated cut-off points for different activity levels for this age group are

based on the vertical axis alone (Trost, Loprinzi, Moore, & Pfeiffer, 2011), only the vertical axis was used for this study. Accelerometers are recognized to provide acceptable validity and reliability for measuring student's activity levels and energy expenditure (Rowlands, 2007).

Student's location was measured using QStarz BT-Q1000x GPS trackers. The GPS trackers were configured to log data every 15 s to be able to record data for seven days. The Qstarz GPS trackers have a manufacturer reported positional accuracy error of maximal three meters and have shown relatively high accuracy and good inter-unit reliability compared to other units (Duncan et al., 2013). The median dynamic error (i.e. the error while moving with different modes of transport, in different urban environments) of the Qstarz GPS tracker was reported to be 2.9 m (Schipperijn et al., 2014).

2.3. Context

In Denmark all public schools are free of charge and mandatory for children aged 6–16. Students do not wear uniforms and each class is gender-mixed with a maximum of 28 students. Public schools in Denmark are typically organized in three tiers: Junior represent grade 0–3, (6–9 years old), middle; grade 4–6, (10–12 years old) and senior; grade 7–10, (13–16 years old) (Pawlowski, Tjørnhøj-Thomsen, Schipperijn, & Troelsen, 2014).

Grade 5–8 participating in this study attends school for an average of 27 h a week. This includes approximately 60 min of recess per day, distributed over two to three breaks. Lunch recess is in general the longest break, lasting 25–30 min (Ministry of Education, 2013). Morning recess and lunch breaks are offered in almost every Danish school, which gives the students the opportunity to go outside. Recess is typically characterized by free play without any organized curriculum.

2.4. Procedures

The participants were asked to wear the accelerometer and GPS for seven consecutive days, though for this study only school day data were used. The monitors were worn by the participants in an adjustable elastic belt around the waist. The participants were instructed to put the belt on as soon as they woke up and to take it off just before going to bed, and to remember to charge the GPS every evening. Instructions were also given regarding removal of the belt during water-based activity or contact sports where the risk of injury or monitor damage was high.

Participants were asked to fill out a short daily diary containing short questions about their day (when they woke up, went to school, reason for not going, ended their school day, went to bed and reasons for not wearing the monitors).

Short text messages were sent out every morning and evening reminding the participants to follow the instruction given. The participants were encouraged to ignore the monitors while wearing them and go about their normal activities during the monitoring period.

Furthermore all four schools provided detailed class time tables for the data collection period.

2.5. Measurement of schoolyard characteristics

In this study a schoolyard is defined as the entire outdoor area of a school. All four schools participating in the study can be categorized as city schools. Two of the schools are located in a densely developed area and have schoolyards about half the sizes of the two others, as shown in Fig. 2 and described in Fig. 1.

All schoolyards were mapped in the field using a high precision GPS (Trimble GeoExplorer XT) and ArcGIS Mobile software. This GPS has a reported maximal positional error of one meter, in

practice the positional error was usually less than 30 cm. The actual mapping was done by preparing one GIS data layer for each potential type of schoolyard feature in advance, and when out in the field, the GPS recording function was activated before walking the outline of each schoolyard feature. The environmental features of the schoolyard were categorized into five main area types before analyses (see Fig. 2).

Fig. 2 shows the distribution of the area types in percent out of the total area per school and the distribution of the area types on each of the four schoolyards.

Each area type was categorized by its primary characteristic. *Grass* represents various play fields and lawn areas, often used for soccer but without any markings or goals. *Multi-court* represents framed play areas on surfaces like artificial grass, rubber, concrete pavers or asphalt designed for different ballgames like basketball and soccer. *Solid surface* represents all paved areas with surfaces like asphalt or concrete pavers. This area type is characterized by level (flat) open areas, often with various painted markings for games and benches placed in different places. *Natural* represents areas with shrubs, trees, natural stones, etc. *Playground* represents areas with playground equipment such as swings, slides and climbing frames, typically grouped and placed on surfaces like sand or gravel. The few areas that could not be categorized into one of these five groups were assigned to the area type *other*, which for instance contains shelters and pent roofs.

The relative share and absolute size of the five area types differs between the four schools as shown in Fig. 1. In general, the schoolyards have flat surfaces and are mostly covered by *solid surfaces* like asphalt. The school with the largest schoolyard stands out because of the large percentage of *grass* compared to the three other schools.

2.6. Data reduction

At the end of the data collection periods the data were downloaded using ActiLife v.4.4.1 and GPS data logger software BT747 (www.bt747.org). All accelerometer and GPS files were processed using the Personal Activity and Location Measurement System (PALMS) to match the two types of data and calculate wear time and PA intensities (Demchak, Kerr, Raab, Patrick, & Krüger, 2011). The combined accelerometer and GPS data were then downloaded into a PostgreSQL database and combined with data from the diaries, class timetables and geographic data. The database was used to select school time for each participant based on the diaries and the class time tables. If there were any individually noted times in the diary, these times were used to determine the school day start and end, if there was nothing noted in the diary, class timetables were used. The diaries indicated that participants in the same class could have different lengths of the same school day. Using individually adapted school start and end times helped to specify in which segment of the school day the behavior in the schoolyard were situated.

In line with international recommendations for accelerometer measurements, continuous periods of 60 min of zero values were classified as accelerometer non-wear time and were removed from the data (Corder, Ekelund, Steele, Wareham, & Brage, 2008; Troiano et al., 2008). Removing non-wear time from the dataset is necessary as sedentary time and non-wear both result in zero values, and identifying what was 'sitting or standing still' and what was accelerometer removed (non-wear), e.g. during night time, is crucial in determining activity levels. Cut points reported by Evenson, Catellier, Gill, Ondrak, and McMurray (2008) have shown the best validity among children and adolescents at every activity level (Trost et al., 2011) and were used to identify MVPA (>2296 counts per minute) and sedentary behavior (<100 counts per minute). Data from the first day of monitoring were removed from the dataset as the instruments were fitted at different times (between 8am and

Schools	Schoolyard description
1	Relatively spacious schoolyard with a large lawn area, some smaller playground areas, a large solid surface area with some markings. Finally, there is a small bush area.
2	Small schoolyard primarily covered by a solid surface with many different markings. A large climbing frame as well as two multi-courts are also present
3	Varied schoolyard with five multi-courts, 3 of which with artificial grass, and a large solid surface area with markings. A playground area with half-shelters is also present.
4	Schoolyard dominated by a large multi-court, and a large solid surface area. In the schoolyard edges playground equipment is present



Fig. 1. Description of the four schoolyards and examples of schoolyard elements.

2pm), leaving the students with unequal opportunities to obtain data from both the morning and lunch recess. Additionally, the first day was removed to minimize the potential effect of the novelty of wearing an activity monitor on children's PA level.

Participants were included in the analysis if they had data (i.e. no non-wear) from the monitors in both the morning recess and lunch break. Absolute and proportional total time, MVPA time, and sedentary time as were calculate for each day, school day, combined recess and lunch breaks, for the total schoolyard and for each of the schoolyard area types. The proportion of time spent in MVPA and sedentary out of the total time spent in each schoolyard area type was calculated as the outcome. This measure indicates with how much MVPA each minute spent in an area type is associated. From a schoolyard design and planning perspective, a higher share of MVPA for one area type compared to another could indicate that providing more areas with a high share of MVPA time could increase the overall activity level in a schoolyard.

Grade was used as a proxy for age and dichotomized into grade 5–6 versus grade 7–8 to approximate the life stage 'childhood' and 'adolescence'.

2.7. Statistical analysis

The Spearman–Brown prophecy formula was used to determine the number of days required to achieve a reasonable reliability of 0.7 (Nunnally, 1975). The Spearman–Brown prophecy formula uses the interclass correlation coefficient (ICC) as a measure of reliability. The ICC is defined as the ratio of between-individual-variance to the sum of the between- and within-individual-variance (Streiner, 1995). A large within-individual-variation requires more days of

monitoring. The ICC was calculated in STATA 12.0 based on the total schoolyard time per day and the number of days required was calculated afterwards using the following formula:

$$N = \left[\frac{ICC_t}{(1 - ICC_t)} \right] \left[\frac{(1 - ICC_s)}{ICC_s} \right]$$

where N is the number of days required for an ICC of 0.7, ICC_t is 0.7, ICC_s is the single day ICC.

Descriptive statistical analyses were carried out in STATA 12.0 to test for significant differences between boys and girls, and children (grade 5–6) and adolescents (grade 7–8) in the use, total time spent and the proportion of time in MVPA, and the proportion of time spent sedentary out of the total time spent in each area type.

Means and standard deviations were calculated for the continuous outcome variables and the differences between them were tested using t -tests and analysis of variance (ANOVA). Significance was set at $p < 0.05$.

3. Results

3.1. The participants and their overall physical activity

Out of the 523 students who consented to participate in the study, valid accelerometer and GPS data were obtained from 509 children. These 509 students had one to four days available for the reliability analyses. The Spearman–Brown analysis revealed that 1.6 days of monitoring were required to achieve a reasonable reliability of 0.7. Data was considered valid if the participants had at least two days of monitoring with both morning and lunch recess. The final sample consisted of 316 participants (46.8% boys) with

two or more days of valid data, leading to a total of 892 days and 1784 individual recess periods that were included in the analyses.

Table 1 shows the daily average wear time, time spent in MVPA and time spent sedentary for all participants, for the total day, school day, recess and schoolyard. The average wear time was 13.2 h per day (SD 2.4), with an average of 59 min spent in MVPA and (SD 27.9) 7.5 h (SD 1.9) sedentary. During the school hours 35% of the daily MVPA and 45% of the daily sedentary behavior were generated.

The average total recess duration was 55 min (SD 7.2) per day, in which the participants spent a mean of 7.8 min in MVPA (SD 4.9). For 47% (25.9 min (SD 8.8)) of recess time participants were engaged in sedentary behavior. Significant differences in recess MVPA were found between gender with boys accumulating 3.1 additional minutes of recess MVPA per day compared to girls ($p < 0.001$). Differences between children and adolescents were present with children accumulating 2.6 additional minutes of recess MVPA per day compared to adolescents ($p < 0.001$). Significant gender differences were also found in the patterns of sedentary behavior with girls gathering 7.5 additional minutes of sedentary time during recess compared to boys ($p < 0.001$). Adolescents accumulated 6.1 additional minutes of sedentary behavior compared to children ($p < 0.001$).

3.2. Physical activity in the schoolyard

On a daily basis, participants spent 46% of recess time in their schoolyard with a mean of 3.9 min in MVPA (SD 4.1). No significant gender difference was found in total schoolyard time. However, boys accumulated 1.9 additional minutes of MVPA time compared to girls ($p < 0.001$) in the schoolyard. Children spent on average 10.5 min more in their schoolyard, engaging in 4.3 additional minutes of MVPA compared to adolescents ($p < 0.001$). The proportion of time spent in MVPA in the schoolyard out of the total recess

MVPA differed between age groups, with adolescents spending 23% and children 64% of their recess MVPA in the schoolyard. The average daily sedentary time in the schoolyard was equivalent to 45% of time spent in the schoolyard. Girls were spending 3.0 min more being sedentary in the schoolyard compared to boys ($p < 0.001$). No significant difference in sedentary time in the schoolyard was found between children and adolescents.

3.3. Time spent in the different schoolyard area types

Table 2 shows the average time spent per day in each area type and the proportion of that time spent in MVPA and being sedentary, respectively. Since one area type, e.g. *solid surface*, can consist of several separate areas in one schoolyard we have chosen not to standardize by square meter per area type as participants could, in theory, spent all the time on one small area of *solid surface* while the total solid surface area is much larger.

Results from the area type *Other* will not be presented further as both the area and time spent here was very limited.

Most time was spent in *solid surface* areas with an average of 16.4 min (SD 8.4), and these areas also attracted the highest average number of participants. An average of 5.7 min (SD 6.8) and 4.1 min (SD 5.0) were spent in *multi-court* and *grass* areas, respectively. Distinct differences between children and adolescents were found in time spent in every area type except *natural*. Compared to adolescents, children spent about 2–3 additional minutes more in the *grass*, *solid surface* and *multi-court* areas. In terms of number of users per area type, the percentage of children present in the different area types was higher than the percentage adolescents in all five area types, with significantly more children in *playground*, *natural* and *grass* area types. However, no significant gender differences were found in time spent in each of the five area types.

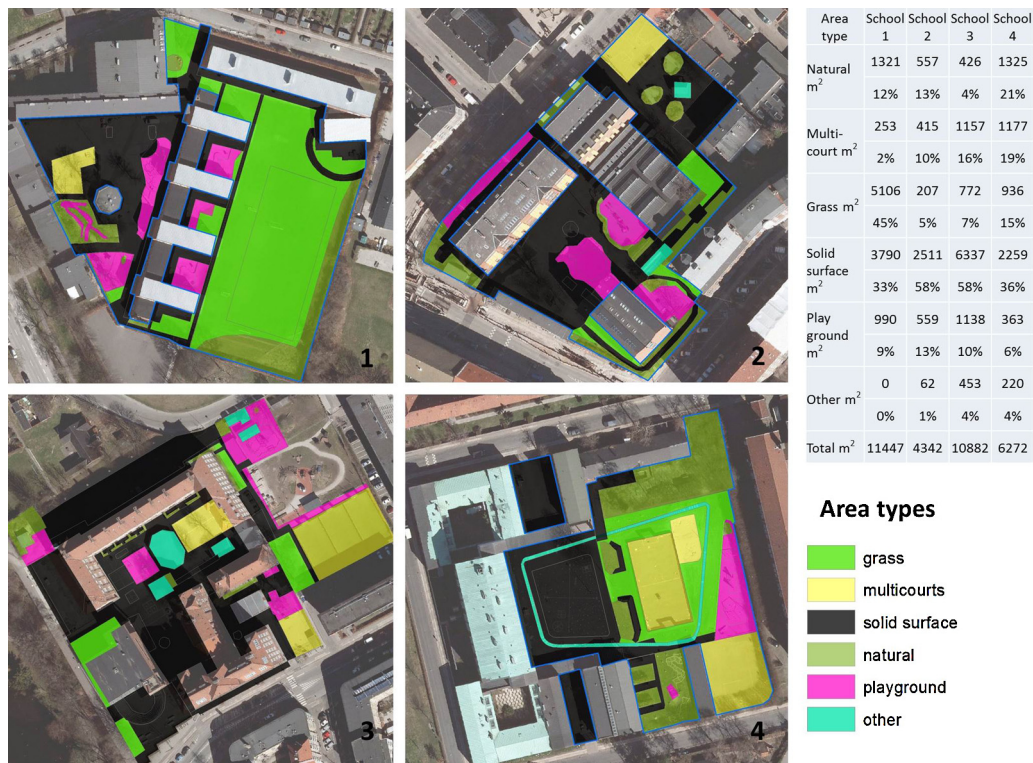


Fig. 2. The distribution of the five area types for each of the four schoolyards, illustrated with schoolyard maps and a table with square meters and the percent of the total area per school.

Table 1
Average daily wear time, time spent in MVPA and time spent sedentary for all participants, for the total day, school day, recess and schoolyard during recess, in minutes. *p*-Values for gender and age-group differences are included, as well as the range and standard deviation of the averages.

	Boys <i>n</i> = 148	Girls <i>n</i> = 168	<i>p</i> -Value	Child <i>n</i> = 175	Adolescent <i>n</i> = 141	<i>p</i> -Value	All <i>n</i> = 316	Min.	Max.	SD
Day										
Time	767.9	810.1	0.01	790.5	790.1	0.98	790.1	317.9	1240.5	144.1
MVPA	68.0	51.1	<0.001	63.9	53.0	0.00	59.0	10.9	148.3	27.9
Sedentary	424.2	469.2	<0.001	434.6	464.9	0.02	448.1	154.4	850.0	111.0
School										
Time	310.6	321.0	0.02	314.5	318.2	0.41	316.1	182.9	399.8	39.6
MVPA	27.1	19.4	<0.001	26.5	18.7	<0.001	23.0	2.2	59.4	11.9
Sedentary	178.5	205.3	<0.001	182.0	206.0	<0.001	192.7	77.3	306.8	40.9
Recess										
Time	54.4	55.6	0.16	55.0	55.0	0.99	55.0	22.4	74.5	7.2
MVPA	9.5	6.4	<0.001	9.0	6.4	<0.001	7.8	0.1	23.2	4.9
SED	21.9	29.4	<0.001	23.2	29.3	<0.001	25.9	3	53.1	8.8
Schoolyard										
Time	25.9	25.3	0.69	30.3	19.8	<0.001	25.6	0	61.1	13.0
MVPA	4.9	3.0	<0.001	5.8	1.5	<0.001	3.9	0	19.9	4.1
Sedentary	9.7	12.7	<0.001	11.3	11.3	0.94	11.3	0	38.1	7.0

3.4. Proportion of time spent in MVPA in the different schoolyard area types

The proportion of time spent in MVPA in each of the area types ranged from 12 to 27 percent, with boys spending a significantly higher proportion (15–32%) of time in MVPA compared to girls (9–23%), and children spending a significantly higher proportion (17–29%) of time in MVPA compared to adolescents (0–25%). *Grass* had the highest proportion of time in MVPA (27%), followed by the *playground* (26%) and *multi-court* (22%) areas. *Grass* and *playground* were the areas with the highest percentage of MVPA for both boys and girls. Boys engaged in MVPA 31% percent of the time in *grass* and 32% of the time in *playground*, while girls engaged in MVPA for 23% of the time in *grass* and 20% of the time in *playground*. The greatest difference between genders was found in *multi-court* areas, with

boys engaging in MVPA for 28% of the time and girls 16% ($p < 0.001$). Fig. 3 shows an example of one boy and one girl's movement pattern and activity level during one recess period, representing a common pattern that boys are more PA than girls.

Grass had the highest proportion of time in MVPA for both children and adolescents. The greatest difference between children and adolescents was found in *multi-court* areas, with adolescents only engaging in MVPA for 7% of the time, compared to 17% for children.

3.5. Proportion of time spent sedentary in the different schoolyard area types

Overall the area type *solid surface* had the highest proportion of time spent sedentary (47%), followed by *natural* (38%) and *multi-court* (33%) areas. Gender differences were found for sedentary

Table 2
Average time spent in each area type in minutes, and the proportion of that time that was spent in MVPA and sedentary, respectively. *p*-Values for gender and age-group differences are included, as well as the range and standard deviation of the averages.

	Boys	Girls	<i>p</i> -Value	Children	Adolescents	<i>p</i> -Value	All	Min.	Max.	SD
Grass										
<i>n</i>	128	126		159	95		254			
Time (min)	4.3	3.8	0.417	5.1	2.2	<0.001	4.1	0.3	25.8	5.0
MVPA	31%	23%	0.02	29%	25%	0.29	27%	0%	100%	28%
Sedentary	25%	37%	<0.001	27%	36%	0.03	31%	0%	100%	30%
Multi-court										
<i>n</i>	135	121		150	106		256			
Time (min)	6.4	5.1	0.146	6.6	4.4	0.011	5.7	0.3	50.3	6.8
MVPA	28%	16%	<0.001	27%	15%	<0.001	22%	0%	100%	26%
Sedentary	25%	41%	<0.001	27%	42%	<0.001	33%	0%	100%	29%
Solid surface										
<i>n</i>	147	167		175	139		314			
Time (min)	16.0	16.8	0.377	17.6	14.9	0.006	16.4	0.3	35.1	8.4
MVPA	15%	9%	<0.001	17%	7%	<0.001	12%	0%	64%	10%
Sedentary	40%	53%	<0.001	41%	55%	<0.001	47%	0%	100%	19%
Natural										
<i>n</i>	117	126		161	82		243			
Time (min)	1.8	1.5	0.153	1.5	1.8	0.280	1.6	0.3	15.5	1.7
MVPA	21%	15%	0.03	21%	12%	0.01	18%	0%	100%	23%
Sedentary	30%	45%	<0.001	36%	41%	0.26	38%	0%	100%	29%
Playground										
<i>n</i>	107	109		156	60		216			
Time (min)	2.3	2.2	0.68	2.6	1.4	<0.001	2.3	0.3	15.8	2.5
MVPA	32%	20%	<0.001	28%	18%	0.01	26%	0%	100%	25%
Sedentary	25%	37%	<0.001	26%	43%	0.00	26%	0%	100%	31%



Fig. 3. An example of one boy and one girl's movement pattern and activity level during one recess period.

behavior in all area types with girls spending significantly more time sedentary compared to boys. *Solid surface* was the area type with the highest proportion of time spent sedentary for both boys and girls, 40% for boys and 53% for girls, respectively. The proportion of time spent sedentary differed slightly between children and adolescents, but significant differences were only found for *multi-court* and *solid surface* where adolescents generated more sedentary time compared to children.

The proportion of time spent sedentary differed in all area types between the two ages groups with adolescents displaying the highest proportion of sedentary time compared to children. Only in the *natural* areas the difference was not significant.

4. Discussion

This study set out to investigate the use of different schoolyard areas and determine if these settings were related to differences in PA and sedentary behavior in schoolchildren attending grade 5–8 using GPS and accelerometers. Significant differences in PA levels in the different area types were found. *Grass* and *playground* areas had the highest proportion of time in MVPA and *solid surface* areas had the highest proportion of time spent sedentary. Boys spent a higher proportion of time in MVPA than girls and adolescents, respectively. The same pattern was found between children and adolescents, respectively. Girls accumulated more sedentary time in all area types compared to boys.

4.1. Time spent in the different schoolyard areas

The participants were in their schoolyard for an average of 25.6 min each day, 46% of their total recess time. The current study found no gender differences in time spent in the schoolyard and in time spent in the different schoolyard area types. This contrasts findings of Fjortoft et al. (2009), who found gender differences in how 6-year-old children used their schoolyard during recess, using objective measurements. However, this might be explained by the fact that the participants included in the study by Fjortoft et al. (2009) were younger than the participants in the current study, and that the number of participants and recess periods included in their study was considerably smaller (71 children, with one recess period of 40 min each). However, results from previous work using direct observation also indicate gender differences in time spent in different types of schoolyard areas (Anthamatten, Brink et al., 2014; Dymont, Bell, & Lucas, 2009).

Most time was spent in the *solid surface* areas. The observational study by Dymont et al. (2009) similarly reported that students spent a large part of their recess time in *solid surface* and *multi-court* areas.

The current study revealed differences between the average time adolescents (19.8 min) and children (30.3 min) spent in their schoolyard. This difference does not seem to be reported by other studies, which is potentially due to the fact that other studies focused on either children or adolescents, rather than both groups. The difference might be explained by the fact that students from grade seven and up (adolescents) were allowed to leave the schoolyard during recess at all four schools in this study.

4.2. MVPA in the different schoolyard areas

Grass generated the highest percentage of time spent in MVPA for all participants in the current study, which indicates its potential to encourage MVPA, irrespective of gender and age. The findings of Martin, Bremner, Salmon, Rosenberg, and Giles-Corti (2012) also indicated that grassed areas were associated with a high level of recess MVPA. The *playground* areas also provided a high proportion of time spent in MVPA, which is comparable with previous findings that equipment such as slides and jungle gyms were associated with a high level of MVPA (Anthamatten, Brink et al., 2014; Dymont et al., 2009). The age differences found in the present study indicates that the relation between *playground* areas and time spent in MVPA is more visible for children than for adolescents.

In the current study there were gender differences in the proportion of time spent in MVPA and time spent sedentary in *grass* and *multi-court* areas, which may be explained by the finding of Dymont et al. (2009). They observed that girls present in areas similar to the *multi-court* area in our study often engaged in sedentary behavior, standing at the periphery of the courts. Blatchford et al. (2003) indicated that boys were significantly more likely to be involved in ball games, and girls more in conversation, sedentary play, jump skipping and verbal games. Paechter and Clark (2007) suggested that PA in the schoolyard was affected by the social expectations among children that boys should be active in competitive sports, which might also explain why girls in our and previous studies were found to be less engaged in MVPA and more in sedentary behavior in the schoolyard compared to boys (Bailey et al., 2012; Hilland et al., 2011; Ridgers et al., 2005; Stratton & Mullan, 2005). Despite spending a similar amount of time in the schoolyard, and the different schoolyard areas, girls may not perceive the facilities and areas as accessible or as enjoyable as boys do. Two reviews exploring the mechanism of participating in PA found that girls experience a wide range of barriers to participate in PA, such as not feeling competent enough, negative reactions from peers about skills, dislike of competitive activities and lack of relevant facilities (Allender, Cowburn, & Foster, 2006; Rees et al., 2006). In the study by Janssen, Twisk, Toussaint, van Mechelen, and Verhagen (2013), evaluating the effectiveness of the intervention program PLAY-grounds on increasing PA, the strongest effect was found for girls aged 10–12. The components especially appealing to girls, e.g. a designated skipping and dance area, increased the PA level in girls from sedentary to moderate.

The difference in the proportion of time spent in MVPA and time spent sedentary between children and adolescents in *solid surface* areas may indicate possibilities for encouraging MVPA. The difference between genders in *solid surface* areas may be explained by boys being involved in competitive games and girls talking on the benches. Dymont et al. (2009) found the highest percentage of students engaged in moderate PA in green areas, similar to the *natural* area type in this study. The reason for not finding a similar result in the current study could be the very limited size of the *natural* areas in the schoolyards included in this study.

Previous findings stated that the schoolyard area per student was positively associated with MVPA during recess (Fairclough et al., 2012; Janssen et al., 2013). In the current study all classes (also the ones not participating in the study) had recess at the same time, and the total schoolyard areas were relatively small, which might have limited MVPA.

In the study by Janssen et al. (2013) distribution of play equipment for each class was a positive predictor for moderate activity in recess. Willenberg et al. (2010) reported similar findings, reporting that settings with loose equipment generated more vigorous activity than the settings without. However, the average schoolyard MVPA only accounted for 15% of the schoolyard time. In contrast, the participants in the study by Fjortoft et al. (2009) were engaged in MVPA for about 58% of their schoolyard time, part of which may be accounted for by the potential effect of the novelty of wearing an activity monitor on student's PA level. This potential effect was accounted for in the present study by removing the data from the first day of monitoring.

4.3. Sedentary behavior in the different schoolyard areas

On average 45% of time spent in *solid surface* areas was sedentary, which indicates a potential for increased activity levels in these areas. Changing the design or the facilities in *solid surface* areas could have a great impact on the average PA level, due to the number of users and the time spent in the areas. Several studies found that adding multicolored markings for different activities to *solid surface* areas increased the MVPA in these areas (Janssen et al., 2013; Stratton & Mullan, 2005).

The large proportion of time spent sedentary in the *solid surface areas* raises questions on whether to focus on moderate or vigorous levels of PA in recess, or if lower levels of activity perhaps are more realistic to achieve. Ridgers, Timperio, Crawford, and Salmon (2013) found that bringing sports equipment to school was associated with a higher percentage of light PA during school breaks for adolescents.

4.4. Study strengths and limitations

The main strength of this study is the use of objective measurements of time, location and activity, for a relatively large sample of participants (316) and recess periods (1784). Including the time dimension adds new important aspects toward understanding PA level in the schoolyard. This study is one of the first reporting how much of the recess time students spent in the schoolyard as well as the time spent in different schoolyard area types for more than one age group.

Additionally, the time aspect makes it possible to look at the proportion of time spent in the activity levels of MVPA and sedentary behavior in the schoolyard. These proportions give important information on how students spend their time in the different areas of the schoolyard.

In most previous studies utility and PA levels were measured by number of students and the percentage of students participating in different PA levels in specific areas in the schoolyard (Anthamatten, Fiene et al., 2014; Dymont et al., 2009; Willenberg et al., 2010).

Another strength of the study was the use of a custom-built database that enabled for inclusion of data from personal diaries and school timetables on an individual and day-by-day basis. This made it possible to determine in which segment of the day the data were generated with high accuracy. If we would not have accounted for individual differences, students that ended school earlier but stayed in the schoolyard would be included in the analysis as having recess time and schoolyard time. As there seems to be a change in behavior and PA in the leisure time compared to school time (Fairclough et al., 2012), this could have generated an unreliable picture of the time spent and PA level in schoolyard.

Several limitations of the study need to be mentioned as well. Accelerometers, also the ones only using the vertical axis, are in general considered valid activity level monitors, but waist-worn accelerometers underestimate the energy expenditure of activities in which the body center of gravity (the waist) does not accelerate or decelerate much, e.g. cycling. However, due to the limited size of the schoolyards and the frequent accelerations and decelerations during most schoolyard activities, we expect that this problem was limited in our study. Using data from all three accelerometer axis, or using multiple accelerometers, e.g. waist and thigh mounted, in future studies might improve activity recognition.

GPS trackers can have difficulties accurately determining locations close to buildings due to poor satellite signal reception, and even in good conditions, they can have an inaccuracy of up to 3 m. Consequently some GPS points could have been assigned to an adjacent area type, or even been placed outside the schoolyard. The schools participating in our study were classified as city schools, and it is likely that the size and design of the schoolyards may vary from schools located in more rural areas, which limits the generalizability of the results. Also the cultural and geographic context of schools and schoolyards is very specific to each setting, which limits the generalizability of the results of this type of study. Dividing the schoolyard in different area types was inspired by area types used in observational studies and facilitated comparison of the results (Anthamatten, Brink et al., 2014; Dymont et al., 2009). However, there were considerable differences within the area types, and a more detailed division might be useful in the future.

Calculating the average time spent in each area type and the proportion of time spent in MVPA and sedentary gives insight in how the participants used their time in schoolyard, but some details of the behavior patterns are missing. Catching the actual behavior of the participants is not possible when calculating averages, and it is not known if the participants move back and forth between areas, or stay for longer periods of time. Using bouts (consecutive periods of time) of MVPA or sedentary behavior as a measure combination with GPS would be preferable in future studies as this will allow for allocation of MVPA and sedentary hotspots, for which the physical characteristics can be described, without a-priori categorizing the schoolyard into different area types. Combining objective measures with direct observation could help estimating which features and games generate bouts of MVPA and sedentary time. Furthermore, exploring the differences between the four schoolyards would be very interesting, analyzing the PA levels in specific areas (i.e. not by area types), and looking at how the elements are placed in the schoolyards. This would create a more in depth understanding of schoolyard PA levels in the exact context. Similarly it would be of relevance to look at the student 'density' in the schoolyard and the different schoolyard areas, both temporal and spatial; i.e. how many students are at same place, at the same time.

5. Conclusion

The current study contributed by adding a temporal dimension to research on schoolyard behavior. Significant differences in PA levels were found between children and adolescents, boys and girls, and between different schoolyard area types. *Grass* and *playground* areas had the highest proportion of time spent in MVPA and *solid surface* areas had the highest proportion of time spent sedentary. Boys and children spent a higher proportion of time in MVPA. Girls accumulated more sedentary time in all area types compared to boys. The gender difference observed in MVPA and sedentary behavior are not surprising, given previous findings. However, it is interesting that no gender difference was found in time spent in each of the areas. This finding emphasizes the importance of investigating the potential of the various settings and features in the schoolyard, in promoting PA among girls. *Grass* and *playground*

areas may play an important role in promoting PA in schoolyards, while improving *solid surface* areas has a great potential for increased PA levels. Future studies could combine GPS, accelerometer and direct observation to determine which types of play are associated with PA behavior.

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